

```
运行 main x
D:\作业\python\作业\taxi_project\.venv\Scripts\python.exe D:\作业\python\作业\taxi_project\main.py
正在加载数据...
数据加载完成, 共 153338 行

===== M1 数据质量报告 =====
缺失值统计:
passenger_count      3525
RatecodeID           3525
store_and_fwd_flag    3525
congestion_surcharge  3525
airport_fee           3525
dtype: int64

异常值统计 (IQR法):
trip_distance 异常值数量: 19439
fare_amount   异常值数量: 16226
M1 特征提取完成

===== M2 开始绘图并保存到 outputs/ =====
M2 图表保存完成

===== M3 需求预测模型 =====
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778066213.289778    21216 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from differ
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778066214.798625    21216 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from differ
随机森林 MAE: 14.23
随机森林 RMSE: 26.43
I0000 00:00:1778066215.235857    21216 cpu.feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI AVX_VNNI_INT8 AVX_NE_CONVERT FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the Ten
神经网络 MAE: 66.39
神经网络 RMSE: 96.17
```

```
===== M4 智能问答系统 =====

请输入问题 (输入 exit 退出): 高峰时段是什么时候?
→ 已为你生成时段需求图: outputs/1_hour_demand.png

请输入问题 (输入 exit 退出): 哪些地方是上车热点?
→ 已为你生成区域TOP10图: outputs/2_top_zone.png

请输入问题 (输入 exit 退出): 车费和距离有什么关系?
→ 已为你生成车费分析图: outputs/3_fare_distance.png

请输入问题 (输入 exit 退出): 哪个模型预测效果更好?
→ 已完成需求预测, 查看MAE/RMSE结果

请输入问题 (输入 exit 退出): 数据里有多少缺失值?
→ 支持查询: 高峰时段、热点区域、车费、需求预测、模型对比

请输入问题 (输入 exit 退出):
```